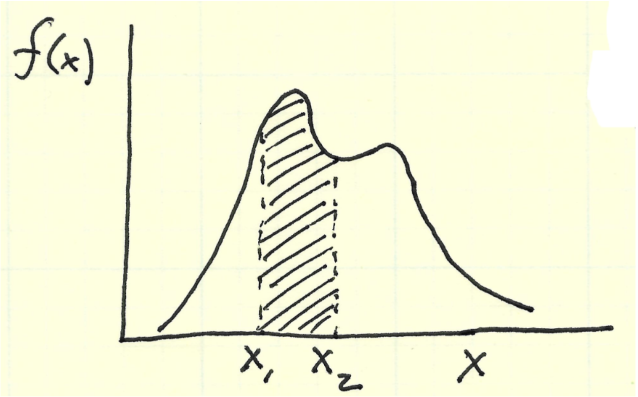
Probability Distributions

Lecture/Lab 04

Physics 4730, Fall 2026



IIDRV's

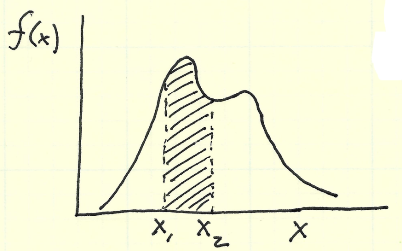
- **Random Variable (RV)** – value results from measurements of a quantity subject to random variations.

- RV types:
  - Discrete
  - Continuous

- **Independent and Identically Distributed (IID)** – variables drawn from the same distribution, that are independent.

  $\longrightarrow$ IIDRV if $p(x_1, x_2) = p(x_1)p(x_2)$

Probability and Probability Density



In a continuous distribution, the probability of measuring any *particular* value is zero! The best we can do is define a "probability density function" (PDF) $f(x)$ such that:

$$\int_{x_1}^{x_2} f(x)dx = P(x_1 < x < x_2)$$

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Descriptive Statistics (location, scale, shape of PDF)

Mean (expectation value): $$\mu = E = \int_{-\infty}^{\infty} x \; f(x) \; dx$$

Variance (standard deviation)^2: $$\sigma^2 = V = \int_{-\infty}^{\infty} (x-\mu)^2 \; f(x) \; dx$$

Skewness (asymmetry): $$\Sigma = E = \int_{-\infty}^{\infty} \left(\frac{x-\mu}{\sigma}\right)^3 \; f(x) \; dx$$

Kurtosis (heavy-tailedness): $$K = \int_{-\infty}^{\infty} \left(\frac{x-\mu}{\sigma}\right)^4 \; f(x) \; dx - 3$$

$\longrightarrow$ "moments" of distribution

(kurtosis of a normal distribution $\equiv 0$)

A couple of other descriptors...

Absolute deviation about d: $$\delta = \int_{-\infty}^{\infty} |x - \mu| \; f(x) \; dx$$

Mode (most probable value): $$\left. \frac{df(x)}{dx} \right|_{x=x_m} = 0$$

Estimates of Descriptors from N Data Points

Mean $$\mu = \int_{-\infty}^{\infty} x\ f(x)\ dx \quad \rightarrow \quad \overline{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

StdDev $$\sigma = \sqrt{\int_{-\infty}^{\infty} (x-\mu)^2\ f(x)\ dx} \quad \rightarrow \quad S = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})^2}$$

Mean Err $$\sigma_{\overline{x}} = \frac{S}{\sqrt{N}}$$

StdDev Err $$\sigma_S = \frac{S}{\sqrt{2(N-1)}}$$

Lab 4: Exponential Distribution

PHYSICS 4730
Lab 04

- Exercises
- Assignment

Exercise 1.

Please review the random variable notebook that the instructor introduced in lecture.

Exercise 2.

Prepare a standalone Python script **expo.py** to perform the following:

- Read data from expodata.dat , assumed to be in the current working directory.
- Have your code determine and write to standard output the parameter β, as inferred from the *mean* and *variance* of the distribution. (You'll have to look up or calculate the mean and variance of an exponential distribution.)
- As a comment in your code, answer the following question: Is it possible to determine the parameter β from the skewness or kurtosis of the distribution? Why or why not?
- Plot the data as a histogram showing the *empirical probability density function*. Superimpose on this the pdf according to scipy.stats, assuming the value for β that you determined above.

Your code should run on the Physics Department Linux Cluster, with the command **python3 expo.py**.

Assignment, due 3 September 2026 *in class*

Problem 1

Submit your code **expo.py** from Exercise 2 above.

```
submit p4730 lab04 expo.py
```

note: Python convention

$$f(x) = \frac{1}{\beta} e^{-x/\beta}$$

versus

$$f(x) = \beta e^{-\beta x}$$

check normalization

$$\int_0^\infty \frac{1}{\beta} e^{-x/\beta} dx = \frac{1}{\beta} \left[-\beta e^{-x/\beta} \right]_0^\infty = 1$$

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